MA∗508 Time Series

Exam-like exercises for L6

Solve the following exercises.

1. Compute the predictor $\hat{x}_4$ for the time series $x_1 = 97$, $x_2 = 102$, $x_3 = 101$ from the stochastic process with $R(k) = 1/(1 + k^2)$. Also compute 95% confidence bounds for x_4.

2. Compute the predictor $\hat{x}_8$ (in August) for the time series x_5, x_6, x_7 of length 3, where $x_5 = 14$ (in May), $x_6 = 11$ (in June), $x_7 = 13$ (in July) from the stochastic process with mean 12 and $R(k) = 1/(1 + 2\sqrt{|k|})$. Also compute the 95% confidence interval for x_8.